
Conduit Quality Assessment Checklist

OR-Ready Reference for GSV · Radial Artery · IMA

How to Use This Checklist

Complete this assessment before handing off any conduit. You are your own quality check — don't wait for the surgeon to find what you should have caught.

- **Visual Inspection**

Assess wall integrity, color, caliber, and any handling injuries before distension.

- **Handling Criteria**

Evaluate how the vessel was managed throughout harvest — spasm, manipulation, thermal risk.

- **Red Flags**

Any item marked here requires verbal disclosure to the surgeon before handoff. No exceptions.

- **Handoff Readiness**

Final pre-handoff checklist. All items must be complete before presenting conduit.

LAMINATE-WORTHY: Print, laminate, and hang in your harvesting suite or keep in your badge holder.

CONDUIT: GREAT SAPHENOUS VEIN (GSV)

VISUAL INSPECTION

- ☐ Vein caliber $\geq$ 3mm at smallest segment
- ☐ Wall color: bluish-white, no discoloration
- ☐ No hematoma or bruising in vessel wall
- ☐ No fenestrations or tears visible
- ☐ Adventitia intact — no wide adventitial stripping
- ☐ Tributaries ligated with appropriate stump length

HANDLING CRITERIA

- ☐ Handled by adventitia only — no direct wall grasping
- ☐ No excessive traction applied during run
- ☐ Tributaries ligated before avulsion (no tear-offs)
- ☐ Cautery kept away from vein wall during dissection
- ☐ Minimal manipulation of vein after harvest

RED FLAGS ☐ DISCLOSE TO SURGEON

- ☐ Hematoma visible in wall segment
- ☐ Fenestration / wall defect present
- ☐ Tributary avulsion with wall involvement
- ☐ Caliber $<$ 2.5mm throughout
- ☐ Thermal injury (brown discoloration) near wall
- ☐ Vein did not distend uniformly
- ☐ Active leak at tributary site after repair attempt

HANDOFF READINESS

- ☐ Distended with heparinized saline — uniform expansion
- ☐ All tributary leaks identified and repaired
- ☐ Length measured and adequate for planned grafts
- ☐ Vein stored in heparinized saline until handoff
- ☐ Ready to verbally present quality to surgeon

NOTES / SURGEON DISCLOSURE:

CONDUIT: RADIAL ARTERY

VISUAL INSPECTION

- ☐ Artery appears pale-pink, soft, and pliable
- ☐ No focal narrowing or spasm bands visible
- ☐ Branch vessels cleanly clipped or ligated
- ☐ No thermal injury discoloration on wall
- ☐ Length adequate for planned anastomosis
- ☐ Allen's test / Doppler confirmed pre-harvest

HANDLING CRITERIA

- ☐ Papaverine applied immediately upon exposure
- ☐ Gentle dissection — no stretching or forceful traction
- ☐ Clips placed perpendicular, not flush to arterial wall
- ☐ No sustained cautery application near artery wall
- ☐ Artery stored in papaverine bath immediately post-harvest

RED FLAGS ☐ DISCLOSE TO SURGEON

- ☐ Vessel tight and non-compressible (spasm)
- ☐ Did not distend under gentle saline pressure
- ☐ Focal narrowing at clip or cautery site
- ☐ Active branch vessel leak not controlled
- ☐ Length insufficient for target anastomosis
- ☐ Allen's test abnormal — not pre-disclosed
- ☐ Thermal injury visible on arterial wall

HANDOFF READINESS

- ☐ Soaking in papaverine solution — fully relaxed
- ☐ Gently distended — assesses uniformly
- ☐ All branch leaks identified and repaired
- ☐ Length confirmed adequate
- ☐ Verbal handoff: spasm status, any repairs made

NOTES / SURGEON DISCLOSURE:

CONDUIT: INTERNAL MAMMARY ARTERY (IMA)

VISUAL INSPECTION

- ☐ Artery appears whitish-pink, soft, pliable
- ☐ Pulsatile — visible or palpable pulsatility
- ☐ Brisk pulsatile flow on division of distal end
- ☐ No kinking along pedicle course
- ☐ Length adequate — reaches LAD without tension
- ☐ No clip placed too close to arterial wall

HANDLING CRITERIA

- ☐ Pedicle retraction was steady, not excessive
- ☐ Intercostal perforators clipped proximal + distal
- ☐ Phrenic nerve not at risk from retraction positioning
- ☐ Cautery used for tissue — not directly on IMA wall
- ☐ Pedicle maintained with adequate soft tissue buffer

RED FLAGS ☐ DISCLOSE TO SURGEON

- ☐ Minimal or sluggish distal flow on division
- ☐ Artery appears pale, tight — possible spasm
- ☐ Kinking or angulation in pedicle course
- ☐ Length insufficient to reach LAD without tension
- ☐ Clip placed at risk of luminal compromise
- ☐ Pedicle injury — hematoma or disruption visible
- ☐ Phrenic nerve at risk from retraction position

HANDOFF READINESS

- ☐ Flow assessed: vigorous, pulsatile stream confirmed
- ☐ Length verified: adequate reach to LAD
- ☐ No kinking in pedicle when heart is decompressed
- ☐ Surgeon informed of any intercostal perforator repairs
- ☐ Verbal handoff: flow quality, length, any concerns

NOTES / SURGEON DISCLOSURE:

Quick Reference: Universal Handoff Standard

Before handing ANY conduit to your surgeon, you should be able to say:

"Here's your [conduit type] — approximately [X] centimeters."

"It looks [good / I had one area of concern which is...]."

"I had [one repair / no repairs] at [location] — holds on distension."

"[Additional conduit] is ready when you need it."

The rule: if you know something the surgeon needs to know about that conduit — say it before they ask.

CASE DATE: _____

CONDUIT(S) USED: _____

QUALITY NOTES: _____
